

Swasthify

by
hack2win

*"Bridging Distances, Enhancing Care
—Empowering Health with AI, IoT,
and Virtual Connectivity"*

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Overview

The Problem

- Chronic diseases (heart disease, cancer, diabetes) cause 71% of global deaths (WHO)
- Nearly 50% of the world lacks access to essential health services
- Rural areas: 45% shortage of healthcare professionals

Access Challenges

- Rural: Long travel, poor infrastructure, high transport costs
- Urban: Overcrowded hospitals, long wait times
- Common issues: Delays, travel hassles, broken care continuity

Tech-Driven Solution

- Powered by AI, IoT, and virtual healthcare platforms
- Enables:
 - Early diagnosis
 - Personalized treatment
 - Real-time medication management

Our Platform

- Bridges patients and doctors, especially in underserved areas
- Reduces travel, improves access, and ensures quality healthcare delivery

Problem Statement

- **Limited healthcare access in both rural and urban areas due to a shortage of healthcare professionals and travel barriers**
- **The need for a virtual consultation platform to connect patients and qualified doctors seamlessly**
- **Rise in chronic diseases like heart disease, skin cancer, and diabetes, requiring early detection and continuous monitoring**
- **Leverage AI-assisted diagnostics for accurate and timely disease detection**
- **Implement digital prescription management for secure and paperless medication records**
- **Integrate an IoT-powered pill dispenser for automatic medication dispensing and adherence tracking**
- **Include an AI-driven medical chatbot to offer real-time health advice and 24/7 support**

Virtual Doctor Consultations

Centralized Database and Role-Based Dashboards

**IoT-based Automated Pill
Dispenser**

Medical summary

**AI-Powered Symptom
Checker**

Personalized Healthcare Tips

**Drug Recommendation &
Treatment Duration (AI Models)**

AI RAG Chatbot – Dr. MediBot

Specific Disease Detection Models

**PrescriptIQ – Prescription
Summarization**

**Idea/
Solution**

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graph TD; A[Idea/Solution] --> B[Virtual Doctor Consultations]; A --> C[Personalized Healthcare Tips]; A --> D[Centralized Database and Role-Based Dashboards]; A --> E[Drug Recommendation & Treatment Duration (AI Models)]; A --> F[IoT-based Automated Pill Dispenser]; A --> G[AI RAG Chatbot – Dr. MediBot]; A --> H[Medical summary]; A --> I[Specific Disease Detection Models]; A --> J[AI-Powered Symptom Checker]; A --> K[PrescriptIQ – Prescription Summarization];
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03

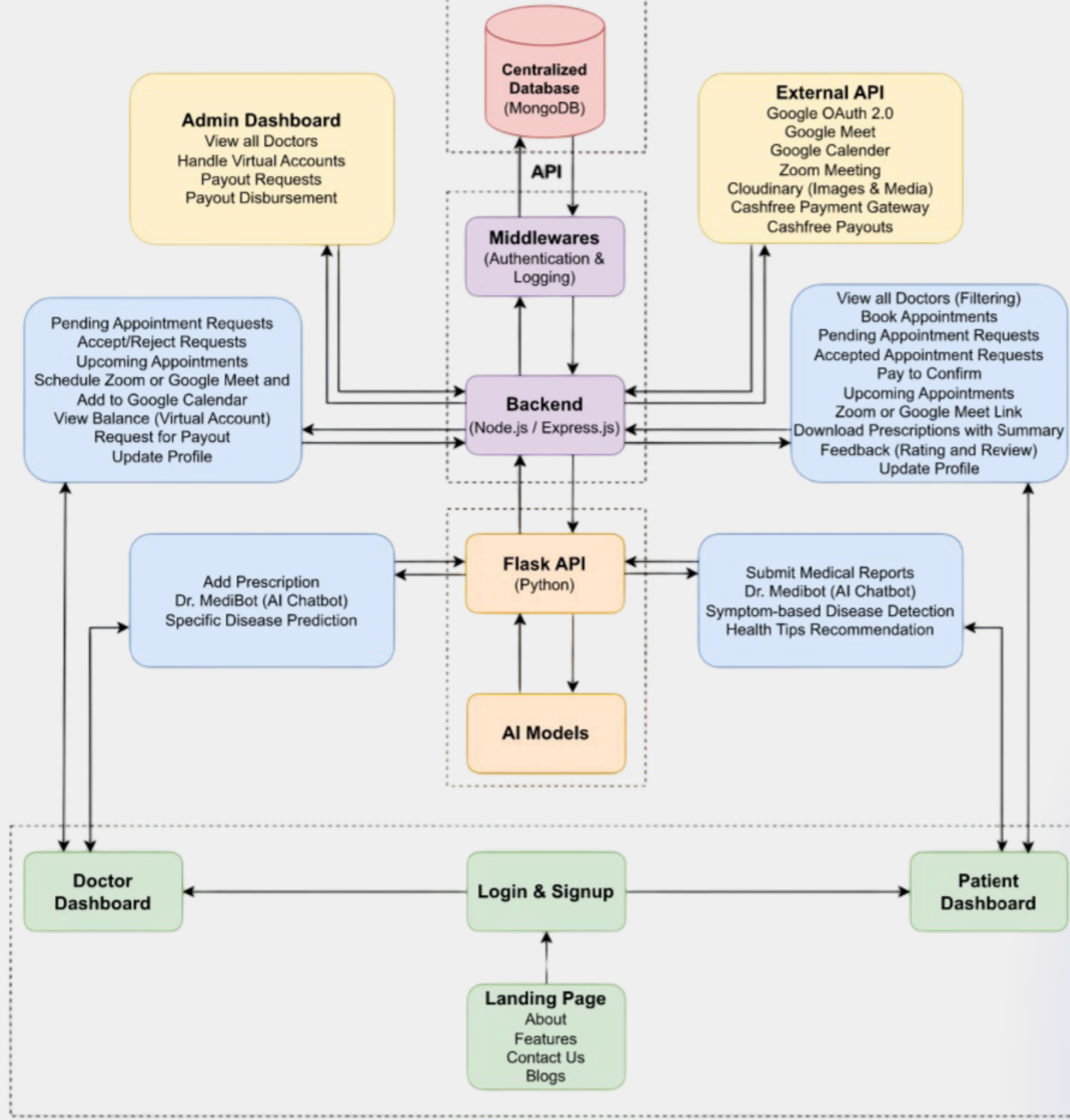
Features

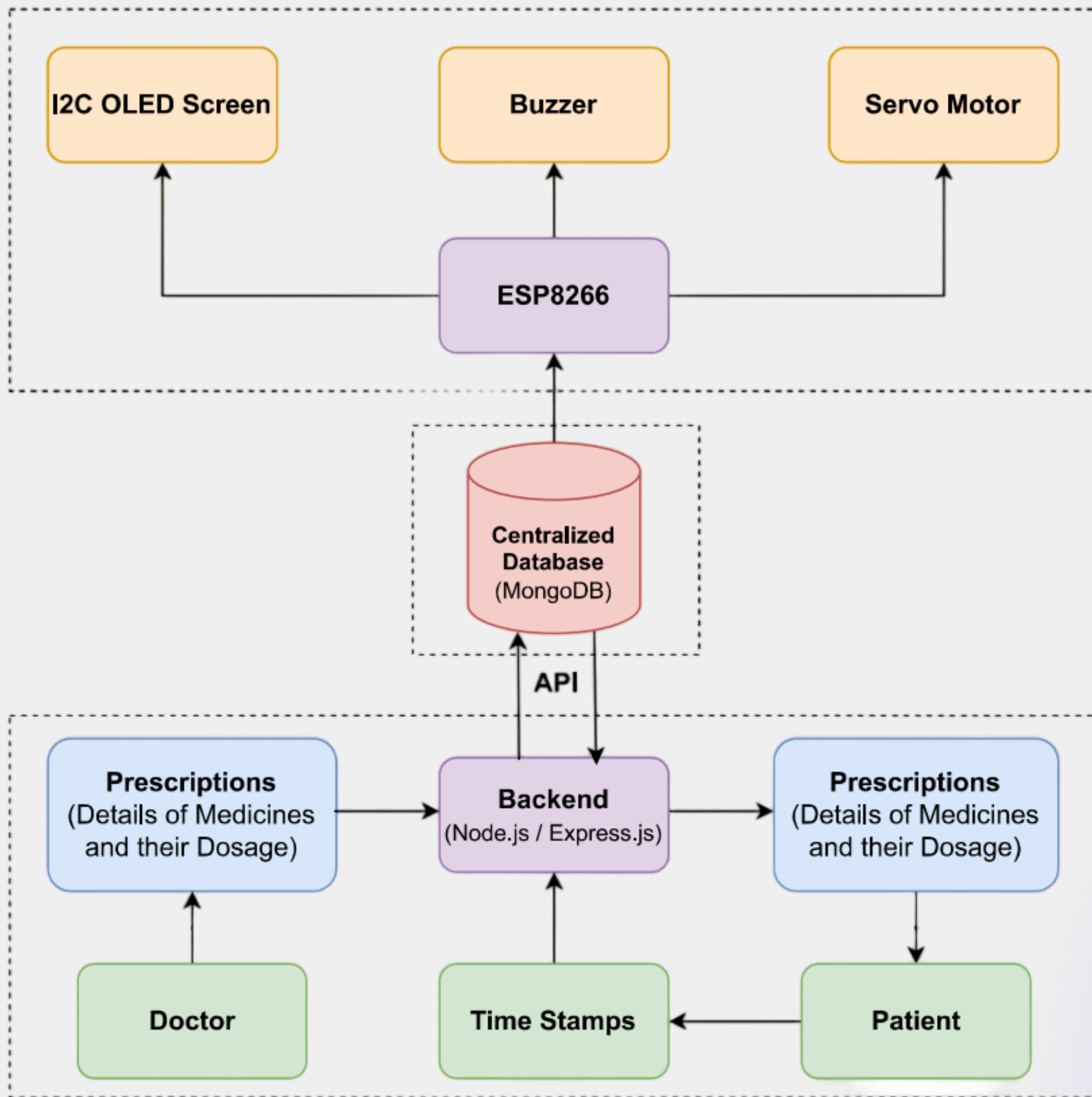
- Doctor-Patient Appointment Management System
- IOT-based Automated Pill Dispenser
- Symptom-based Disease Detection System
- AI Chatbot (Dr. MediBot)
- Personalized Healthcare Recommendations
- Prescription Summarization System (PrescriptIQ)
- Specific Disease Detection Models
- Centralized Database and User Dashboards
- Mini-Escrow System for Secured Payments
- Drug Recommendation and Duration of Treatment



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IOT Flowchart

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Tech stack

Frontend:

- HTML/CSS
- TailwindCSS & ChakraUI
- MaterialUI & Shadcn
- Accertinity & Spline for 3D
- Framer Motion
- JavaScript
- React

Backend:

- Node.js
- Express.js
- Flask (Python)

Authentication & Security:

- JSON Web Tokens
- Google OAuth 2.0

Machine Learning / AI:

- Python
- TensorFlow
- Jupyter Notebook
- LLaMA
- PyTorch
- FAISS

Database:

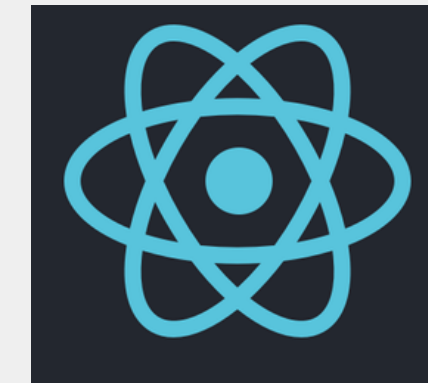
- MongoDB

IoT / Hardware:

- ESP8266 NodeMCU
- Servo Motor
- Buzzer

External APIs

- Cloudinary
- Google Meet



Revenue Model



BASE PLAN

Free account creation

- Appointment Booking using mini-escrow system
- 10 AI queries/month (text/audio/pdf)
- Sync with 1 IoT Pill Dispenser
- In-app notifications for dosage
- Access to basic AI tools:
- Prescription Summarization
- Health Tips Recommender
- Drug Recommender

PATIENT PREMIUM

₹199/month or ₹1,999/year

Best for: Active users & chronic care patients

- Unlimited AI queries
- Smart Health Insights from ML models (CNN, LSTM)
- Multi-format AI access: Text, Audio, PDF
- Multi-device IoT Pill Dispenser sync
- WhatsApp + Email dosage reminders
- Priority Support

DOCTOR PREMIUM

₹499/month or ₹4,999/year

Best for: Practicing doctors & telemedicine clinics

- Advanced Patient Dashboard
- Predictive analytics on patient data
- Featured in search results
- EHR Integration (on request)
- Early access to new AI tools
- Advanced Disease Prediction APIs

■ 01

Advanced AI Models:

Add more AI models for diagnosing a wider range of diseases, making the platform a comprehensive diagnostic tool.

■ 02

Wearable Device Integration:

Sync wearable health devices to monitor real-time health metrics for personalized care and automated doctor alerts.

■ 03

Global Telehealth Expansion:

Offer multi-language support and expand virtual consultations globally to provide access to specialized doctors worldwide.

■ 04

Online Pharmacy & Lab Test Bookings:

Enable users to order medicines online and book home lab tests directly through the website for added convenience.

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Future Scope

Thank You!